

An Efficient Website Fingerprinting for New Websites Emerging Based on Incremental Learning

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Abstract—Website fingerprinting attacks leverage encrypted traffic features to identify specific services accessed by users within anonymity networks such as Tor. Although existing WF methods achieve high accuracy on static datasets using deep learning techniques, they struggle in dynamic environments where anonymous Websites continually evolve. These methods typically require full retraining on composite datasets, resulting in substantial computational and storage burdens, and are particularly vulnerable to classification bias caused by data imbalance and concept drift. To address these challenges, we propose EIL-WF, a dynamic WF framework based on incremental learning that enables efficient adaptation to newly emerging websites without the need for full retraining. EIL-WF incrementally trains lightweight, independent classifiers for new website classes and integrates them through classifier normalization and energy alignment strategies grounded in energy-based model theory, thereby constructing a unified and robust classification model. Comprehensive experiments on two public Tor traffic datasets demonstrate that EIL-WF outperforms existing incremental learning methods by 6.2%–20.2% in identifying new websites and reduces catastrophic forgetting by 5.4%–20%. Notably, EIL-WF exhibits strong resilience against data imbalance and concept drift, maintaining stable classification performance across evolving distributions. Furthermore, EIL-WF decreases training time during model updates by 2–3 orders of magnitude, demonstrating substantial advantages over conventional full retraining paradigms.

Index Terms—Website fingerprinting, Tor anonymous network, traffic analysis, incremental learning.

I. INTRODUCTION

ANONYMOUS communication networks, particularly the Tor network, are widely valued for their strong privacy protection features [1]. Through its distinctive onion routing and multi-hop encryption mechanisms, the Tor network effectively obscures the traceability of user communications,

thereby significantly enhancing privacy security. However, this high level of anonymity remains vulnerable to certain advanced traffic analysis techniques [2], [3], such as Website Fingerprinting (WF) attacks [4], [5].

Websites display distinctive traffic characteristics due to variations in their structure, content, and interaction patterns, collectively termed the fingerprints of websites. The WF methods rely on classification models, where attackers collect traffic from target websites to train models and extract fingerprints, thus constructing a set of monitored websites. As shown in Figure 1, during the deployment phase, attackers only need to analyze the traffic characteristics of users accessing the Tor network to identify the websites they visit.

In recent years, DL-based WF methods have demonstrated remarkable effectiveness in static data classification tasks [6], [7], [8], [9], [10], [11], [12], [13]. State-of-the-art benchmark models like DF [6], TikTok [7], VarCNN [8], WF-Transformer [10] and ARES [12], achieve classification accuracy rates of 98%–99% on both AWF [14] and DF [6] datasets. However, these static classification paradigms struggle to maintain performance in dynamic network environments where new website categories continuously emerge. When new website categories are introduced, as illustrated in Figure 1, the model must be retrained on both old and new data and subsequently deployed. This retraining paradigm incurs significant computational costs and introduces memory challenges from storing large historical datasets, particularly for devices with constrained storage capacity.

Class Incremental Learning (CIL) [15], originally developed in the field of computer vision, has recently gained substantial attention as a promising paradigm for addressing the challenge of dynamically expanding monitored datasets. This approach focuses on building deep learning models with the capability for continual learning, enabling them to incrementally acquire knowledge of new classes from sequential data streams while maintaining recognition performance on previously encountered classes [16], [17]. The concept of CIL offers valuable insights for WF, where the set of monitored websites evolves over time. However, the Tor network's distinctive multi-hop encrypted routing mechanism enhances the anonymity and resistance to traffic analysis, introducing fundamental differences in the feature space between WF and conventional visual domains. Consequently, directly transferring existing CIL techniques to WF scenarios often results in suboptimal performance.

Moreover, applying continual learning to WF in dynamic network environments introduces two challenges:

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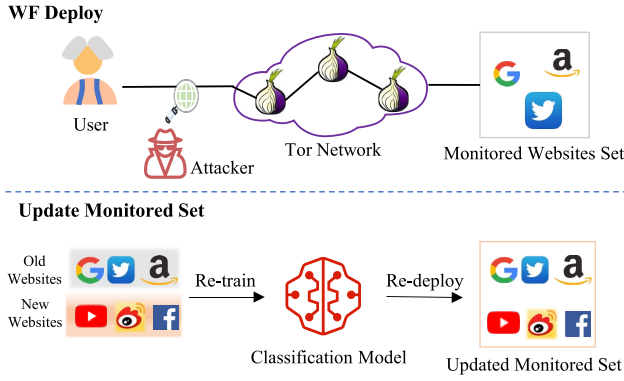


Fig. 1. The illustration of WF deployments, during the update of the monitored set, the saved data of old websites and the data of new websites are combined to retrain the WF classification model, which is then redeployed.

1) Imbalanced training samples: In many stages, training data are collected under favorable conditions, with popular websites and relaxed security policies providing abundant samples. In contrast, strict privacy protections, limited monitoring access, or narrow data collection windows at some stages result in scarce data. This imbalance can cause the model to favor well-represented websites, introducing bias in its predictions. 2) Traffic feature drift: Over time, updates to network protocols, changes in webpage content structures, and adjustments in server response behaviors can all lead to shifts in the traffic feature distribution of a given website. These shifts may lead to inter-class feature space drift, making it difficult for conventional CIL methods to maintain stable recognition performance across different stages.

To address the aforementioned challenges, this paper proposes the EIL-WF framework, a lightweight solution that dynamically updates the monitoring set via stage-specific classifiers, eliminating the need to cache historical data. Initially, a fixed backbone network is trained to extract general traffic features. For each new batch of websites, only the corresponding lightweight classifier is trained, while parameters from earlier stages remain frozen. This architectural separation ensures parameter independence across classifiers. As a result, classifiers trained on low-sample data are shielded from the influence of high-sample distributions, thereby mitigating model bias caused by data imbalance. At the same time, allocating frequently updated website traffic to dedicated classifiers enables targeted local optimization, thereby reducing interference with traditionally stable categories and isolating their feature spaces from contamination.

To support effective classifier selection during inference in a unified model, we introduce free energy scores as the confidence metric, inspired by energy-based modeling theory. This metric provides a more robust and flexible alternative to conventional confidence estimators, especially under the backward-compatible constraints of incremental learning. By assigning lower energy to in-distribution data and higher energy to out-of-distribution data, it enables more accurate and dynamic routing of samples across classifiers. Furthermore, to improve the fairness inference of classifiers, we design two normalization loss functions that normalize the energy output

distribution of classifiers from each phase. By reducing the mutual information between data and non-target classifiers and constraining excessively high energy values of classifiers, we ensure that the energy output levels remain consistent across different stages. In addition, we propose an energy alignment strategy which improves the performance of classifiers trained on low-sample data without introducing additional parameter calculations.

The specific contributions are as follows:

- We propose EIL-WF, an incremental learning framework for website fingerprinting, which trains multiple independent lightweight classifiers. This design enables efficient model updates without requiring access to previously collected data from old websites. Additionally, we introduce a free energy-based confidence metric to quantify the reliability of each classifier.
- We develop a classifier normalization training strategy, incorporating two dedicated regularization loss functions. This strategy ensures both the independence of classifiers and consistently high confidence levels for in-distribution data, effectively improving model stability during incremental updates.
- We design an energy alignment strategy that mitigates the negative impacts of data imbalance and distribution shifts. By aligning the energy outputs of classifiers trained on datasets with varying scales, this approach enhances model robustness in dynamic environments.
- We conduct comprehensive experiments on two widely-used public WF datasets, DF and AWF. The results demonstrate that EIL-WF significantly outperforms existing incremental learning methods in terms of both accuracy and resistance to catastrophic forgetting. Furthermore, we thoroughly evaluate the generalization ability and practical effectiveness of EIL-WF in dynamic website fingerprinting scenarios.

II. RELATED WORK

In this section, we first introduce website fingerprinting, which has been significantly enhanced on static data through the use of deep learning in recent years. We then discuss class incremental learning, a technique that has gained considerable attention in the field of computer vision.

A. Website Fingerprinting Attack

WF attack is a technique that identifies specific websites visited by users through the analysis of patterns in network traffic. It leverages information such as packet size, timing intervals, and traffic direction to identify visited websites, effectively bypassing the protections offered by anonymity networks such as Tor [18]. As network environments become increasingly complex and as Tor's defensive measures are implemented, WF methods continuously evolve and improve.

Earlier WF methods primarily relied on manually configured feature extraction techniques and traditional classifier [3], [19], [20], [21], [22], such as Support Vector Machines [21] and k-Nearest Neighbors [22]. However, the effectiveness of these methods is constrained by feature selection processes and

classifier designs [23]. Their performance often degrades in the presence of dynamic traffic patterns and adversarial defenses deployed in realistic, large-scale network environments.

With the rise of deep learning, WF leveraging deep learning have become a focal point of research. The Deep Fingerprinting (DF) [6] method utilizes one-dimensional (1D) convolutional neural networks to automatically learn and extract traffic features. Notably, it achieves high classification accuracy by solely relying on packet direction features. Subsequent advancements have further enhanced the performance and robustness of WF attacks by incorporating novel deep learning techniques [7], [8], [9], [10], [24]. For instance, such as VarCNN [8] and WF-Transformer [10], enhance classification performance further by incorporating architectures like ResNet, dilated convolutions, and Transformer models. These additions allow the models to handle high-dimensional and complex network traffic more effectively. TikTok [7] and RF [9] models further exploit packet temporal dynamics to generate more robust traffic representations, underscoring the distinctive role of temporal features in WF attacks, particularly when dealing with defensive data, where they exhibit enhanced performance. The Holmes [24] method leverages temporal and spatial traffic patterns to identify websites early in transmission, enhancing attack efficiency and reliability.

To address the identification challenges arising from traffic mixing during multi-tab browsing in the Tor network, [25], [26], [27] proposed a traffic conversion technique based on segmentation point extraction, which segregates multi-tab flows into distinct single-label streams to facilitate classification. However, its classification performance deteriorates markedly as the number of open pages increases. Subsequently, BAPM [28] introduced an attention mechanism that substantially enhances the model's capacity to represent multi-label data, thereby improving identification accuracy. TMWF [11] tackles multi-label WF identification as an ordered set prediction problem, introducing an end-to-end multi-label WF model capable of accurately classifying multi-label network traffic. ARES [12] reframes the problem as a multi-classifier task based on local traffic patterns and incorporates an improved Transformer model (Trans-WF), achieving high recognition accuracy even under unknown label counts and adversarial defense conditions.

Faced with the common issue of limited training samples in real-world network environments, researchers have proposed various few-shot learning-based fingerprinting methods. These approaches maintain high recognition accuracy despite limited training data, effectively reducing dependence on large-scale labeled datasets. For instance, Triplet Fingerprinting (TF) [29] employs a triplet network to achieve N-shot learning, enhancing the model's generalization capability. Contrastive Fingerprinting (CF) [30] adopts a contrastive learning framework combined with data augmentation strategies, demonstrating good accuracy and robustness under data-scarce conditions. NetCLR [13] integrates semi-supervised and self-supervised learning to enhance traffic representation, maintaining strong classification performance even in previously unseen network conditions. However, WF

methods based on few-shot learning models often entail high spatial complexity (e.g., TF with a complexity of $O(n!)$), making them less suitable for large-scale data processing. Furthermore, although methods such as NetCLR can maintain accuracy over time, they still struggle with rapidly changing websites, whose evolving feature space requires retraining on complete datasets.

In summary, although current WF methods have attained notable advancements, several critical challenges persist. Most existing website fingerprinting approaches assume a static monitored set and lack systematic investigation into the dynamic expansion of monitored websites, limiting their adaptability to real-world scenarios where website categories evolve continuously. Additionally, challenges such as model bias due to imbalanced sample distributions and local feature drift in frequently updated websites remain inadequately addressed.

B. Class Incremental Learning

When a deep learning model is trained on new tasks or data, it tends to over-adapt to the new data, resulting in poor performance on previously learned knowledge. This phenomenon is known as catastrophic forgetting. To address this issue, researchers have focused on developing efficient algorithms that enable models to continuously learn new categories from dynamically evolving data streams. These algorithms allow models to incrementally absorb and adapt to new information without requiring complete retraining, thereby significantly reducing computational and storage costs. This emerging learning paradigm is widely referred to as Class Incremental Learning (CIL) [31], [32].

Class incremental learning has made notable strides in the field of computer vision. Depending on the nature of the problem and the applied strategies, current CIL methods can be broadly classified into two categories: data-based methods [15], [33] and parameter-based methods [34], [35]. Data-based CIL methods focus on preserving knowledge of previous categories by storing portions of data or generating representative samples, with the aim of mitigating the forgetting of old categories when new ones are learned [34]. These methods, especially replay-based algorithms, have been widely adopted in image classification tasks due to their simplicity and practicality. However, they typically rely on relatively small sample sets for rehearsal, often much smaller than the size of the training set in each incremental learning phase, potentially causing data imbalance problems. Moreover, the requirement to replay old category samples increases storage overhead and risks that these samples may no longer accurately represent their features over time. This problem is particularly pronounced in network traffic scenarios, where the dynamic nature of data can cause past samples to become misaligned with current conditions.

Parameter-based class incremental learning, on the contrary, focuses on adjusting and optimizing model parameters to retain knowledge of old categories while integrating new ones. One commonly used approach is parameter regularization, which evaluates the importance of model parameters and

constrains their updates to preserve knowledge of previous categories [36]. However, due to the inherent ambiguity and imbalance in network traffic, maintaining the consistent importance of the parameters across different learning stages becomes challenging, complicating the training and deployment of such models in real-world scenarios. Another approach involves dynamically modifying the model structure by expanding or pruning it to incorporate new category knowledge [16], [37], [38]. Zhou et al. proposed EASE [16], which introduces an expandable subspace ensemble, generating task-specific subspaces via adapters and synthesizing historical class features through semantic-guided prototype completion without exemplars. Subspace recombination enables incremental learning in high-dimensional decision space, offering a novel methodological perspective for CIL. Although this method effectively adapts to new categories, it also needs to address potential interference between models at different learning stages.

III. PRELIMINARIES

Although early research achieved significant success in WF within static classification tasks, dynamically updating website classification models has remained a persistent challenge. In this section, we first define a novel WF learning paradigm based on incremental learning. Furthermore, we introduce the energy model theory, which serves as an effective classification confidence evaluation metric for the proposed incremental learning framework.

A. CIL Problem Definition of WF

Both class incremental learning-based WF and traditional WF fundamentally train a classification model to recognize websites in the monitored set. The difference lies in the updating of the monitored set: traditional methods require saving all historical data and retraining the model by replaying this data to ensure that the model can recognize all historical samples [31]. In contrast, the CIL method only necessitates the use of the latest sample data for training, allowing the model to gradually learn new classes or samples without the need to retain all historical data. Below is a brief definition of incremental learning in WF.

Incremental learning for WF constructs an initial deep learning-based classification model M_0 . At each subsequent learning stage $t \in \tau = \{0, 1, \dots, t\}$, the model is updated using only the current monitored subset $D_t = \{(x_i, y_i)\}_{i=1}^{K_t}$, where K_t denotes the number of website classes at stage t and $(x, y) \sim P_D^t$ represents the data distribution at this stage. The updated model M_t is expected to preserve knowledge of prior websites $\bigcup_{i=0}^t D_i$ while minimizing catastrophic forgetting across stages.

In the multi-classifier framework considered in this study, each classifier f_θ^i , where $i \in \tau$ and θ represent the parameters of the classifier, is trained on the data from a specific incremental stage D_i . During inference, for an input sample x , we define it as in-distribution with respect to classifier f_θ^i if $x \in D_i$, meaning the sample conforms to the distribution seen during training of that classifier. Conversely, if $x \notin D_i$, it is considered

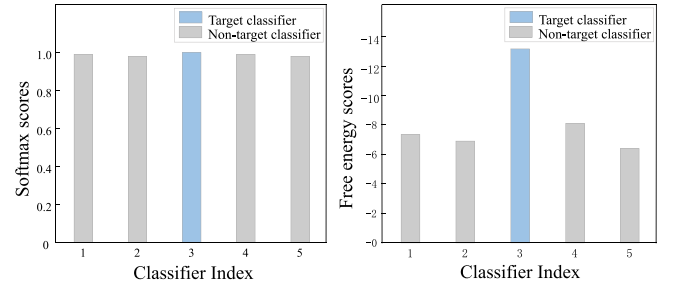


Fig. 2. Illustration of the softmax and free energy scores of sample across target and non-target classifiers.

out-of-distribution relative to f_θ^i . For a given input x , the classifier f_θ^i such that $x \in D_i$ is referred to as the target classifier, and the remaining classifiers f_θ^j for which $x \notin D_j$ are considered non-target classifiers. The central challenge lies in correctly identifying the target classifier for each test sample and effectively distinguishing in-distribution samples from out-of-distribution samples, particularly in the presence of limited training data and evolving decision boundaries.

B. Energy-Based Model Theory

In this method, we use free energy as a confidence scoring criterion to replace the commonly adopted softmax function in WF tasks. Although softmax is computationally efficient, its strong reliance on posterior probabilities often leads to overconfidence when the model encounters out-of-distribution samples [39]. As shown in Figure 2, softmax tends to produce scores close to 1 across multiple classifiers, resulting in a lack of clear separation and reduced effectiveness in selecting the appropriate target classifier during inference. In contrast, the energy-based scoring function does not depend on a shared class probability structure and supports task-wise independent modeling. This property is particularly advantageous for discriminative selection in multi-classifier settings. Free energy provides a meaningful measure of the compatibility between the input and each sub-classifier, enabling dynamic selection and effective inter-class discrimination. Such a mechanism offers greater robustness and discriminative power in incremental scenarios characterized by ambiguous class boundaries and sparse samples. Below, we provide a brief overview of the fundamental principles of energy models.

The foundational concept of the energy model, as proposed by LeCun et al. [40], involves the creation of a function that translates the model's output values into an energy scalar, which indicates the interdependence between the data and the model parameters. This energy scalar, through the application of the Gibbs distribution, can be converted into probability density, thereby enabling a more flexible classification approach for the model.

Firstly, the classifier of a deep learning model can be abstracted as a function f with a value domain in $\mathbb{R}^K$, where K denotes the total number of classes. For a given input data point x , after passing through the neural network, it yields a logits vector. For class $y_i \in \mathbb{R}^K$, the energy score is

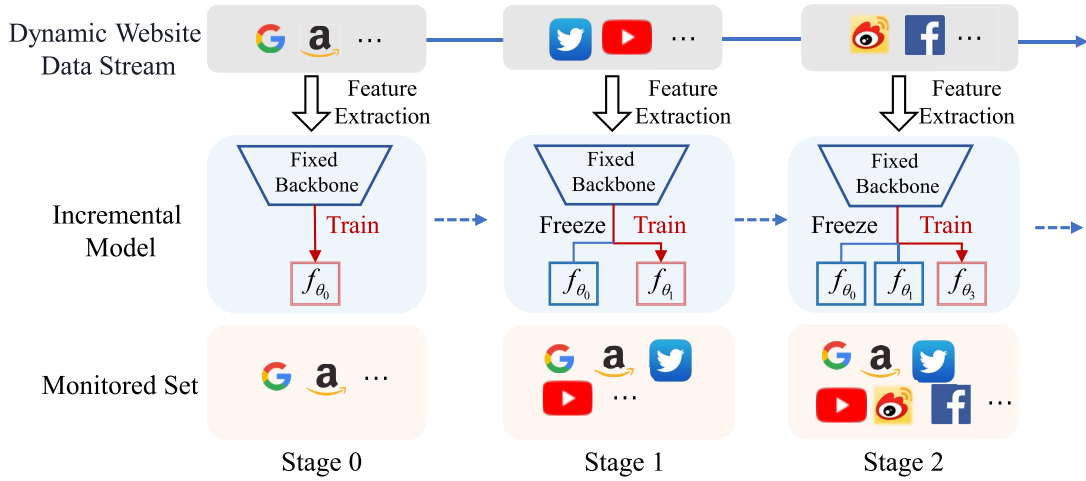


Fig. 3. The illustration of the training phase of EIL-WF. The system incrementally learns from a continuously evolving stream of website traffic data. In the initial stage, a pre-trained and frozen feature extractor is used to train the first classifier for the initial set of monitored websites. In each new stage, a classifier is independently trained for newly added websites, while the backbone and previous classifiers remain fixed to preserve past knowledge. The final model consists of the shared feature extractor and all stage-specific classifiers, each handling data from its respective training stage.

defined as:

$$E(x, y_i) = -f(x)[y_i]. \quad (1)$$

Here, $f(x)$ provides the logits vector, and $-f(x)[y_i]$ represents the logit for class y_i .

Through Gibbs distribution, the energy values can be transformed into probability densities [40]:

$$p(y_i|x) = \frac{e^{-E(x, y_i)/T}}{\sum_{y' \in \mathbb{R}^K} e^{-E(x, y')/T}}, \quad (2)$$

where T is the temperature parameter controlling the smoothness of the distribution; specifically, as $T \rightarrow 1$, the Gibbs distribution approximates the softmax activation values. The denominator serves as the normalization constant, known as the partition function. The negative logarithm of the partition function defines the Helmholtz free energy $F(x, f_\theta)$, which is expressed as:

$$F(x; f) = -T \log \sum_{y \in \mathbb{R}^K} e^{f(x, y)/T}. \quad (3)$$

Unlike softmax confidence scores, which are prone to over-confidence issues, energy scores are theoretically aligned with the input's probability density. A classifier typically assigns lower energy scores to data observed within the training set, while higher energy scores are associated with unseen out-of-distribution data. Additionally, under the same parameters, the energy output of the target classifier should be lower than that of non-target classifiers.

IV. DESIGN OF EIL-WF

In this section, we first present an overview of the proposed EIL-WF. Subsequently, we detail the optimization procedure of our method.

A. Solution Overview

In this work, we propose a general incremental learning framework free of rehearsal for WF, aiming to enable

existing WF methods to have continuous learning capability. The proposed EIL-WF is an incremental learning method that employs the dynamic network structure strategy, and its network structure consists of two main components: the backbone network and the classifier. To simplify the design and effectively extract raw traffic features, we select the well-established *DFnet* as the backbone network for extracting network traffic features. The classifier is composed of fully connected layers using linear activations. Our proposed solution is also suitable for other WF models, such as VarCNN, WF-Transformer, and others.

The training phase of the proposed EIL-WF involves a continuous process of appending classifiers. As illustrated in Figure 3, the proposed EIL-WF trains an isolated classifier for the newly added websites of each stage, updating the monitored set while completing the model update. In the initial stage, a pre-trained backbone network f_η is provided, which is frozen for subsequent stages. The initial model M_0 consists of $\{f_\eta|f_\theta^0\}$, where f_θ^0 denotes the classifier of stage 0. η and θ represent the parameters of the backbone and classifier, respectively. Subsequently, for each incremental stage $i, i \in \tau$, we train a specific classifier f_θ^i to learn the website traffic knowledge P_D^i of that stage. Upon completing t learning stages, the updated model M_t is represented as $\{f_\eta|f_\theta^0, f_\theta^1, \dots, f_\theta^t\}$.

In the testing phase, traffic first passes through the backbone network for feature extraction, and then the extracted features are fed into each classifier. The system selects the classifier with the highest confidence for the test data (Eq. (4)), and the classification result of that classifier is taken as the final prediction result (Eq. (5)).

$$i^* = \arg \max_{i \in \tau} H^i(x), \quad (4)$$

$$y^* = \arg \max_{y \in Y^*} f_\theta^{i^*}(y | x). \quad (5)$$

As shown above, a fair confidence function is essential to ensure that the data is predicted in the correct classifier. The energy score, which is theoretically consistent with probability

density and less susceptible to overconfidence, demonstrates an effective ability to distinguish between in-distribution and out-of-distribution data [34], [39]. However, when applied to website fingerprinting, the inherent homogeneity and ambiguity in the traffic of certain websites compromise the capability of the aforementioned approach to mitigate inter-stage interference. Furthermore, it is prone to the influence of website sample imbalance, resulting in an inability to effectively standardize output levels across different classifiers trained on dynamic website data streams.

To unify the energy output levels across different stages of WF classifiers, we design two regularization loss functions: energy constraining and parameter distancing. These functions are employed to normalize the training of classifiers, thereby mitigating overfitting issues and inter-stage classifier interference. The details of the classifier normalization training are elaborated in Section IV-B. In response to the challenge of website sample imbalance during WF model training, we introduce energy scalars for their corresponding classifiers. This alignment of energy scores with classifiers trained on high-sample website datasets aims to prevent model bias, as further described in Section IV-C. During the testing phase, a fair confidence function is designed to ensure that test traffic is accurately classified by the corresponding classifiers, with the specifics outlined in Section IV-D.

B. Classifier Normalization Training

Similar to traditional classifier training, the multi-classifiers trained by the proposed EIL-WF need to maintain high classification confidence for the websites at their respective stages. To this end, we first employ softmax cross-entropy to optimize the model parameters:

$$\mathcal{L}_{\text{CE}}^{(t)} = \mathbb{E}_{(x,y) \sim P_D^{(t)}} [-\log p(y|x)]. \quad (6)$$

Combining Eq. (2) and Eq. (3) reveals that cross-entropy optimization inherently minimizes free energy $F(x; f_{\theta_t}) = -\log \sum_{y'} e^{f_{\theta_t}(x,y')}$, as shown in Eq. (6) and Eq. (7). From an information-theoretic perspective, cross-entropy loss directly bounds mutual information between data x and labels y through:

$$I(x; y) \geq \log K - \mathcal{L}_{\text{CE}}^{(t)}, \quad (7)$$

where K is the number of categories. Maximizing $I(x; y)$ ensures classifier θ_t focuses on current-stage features.

Unlike traditional approaches, our approach also requires each classifier to exhibit low classification confidence for data from non-target stages. The energy model theory requires $F_{\text{in-distribution}} \ll F_{\text{out-of-distribution}}$. However, the emerging of new websites data causes distribution non-stationarity and class boundary blurring, leading to two critical issues: 1) Excessive sharpness in prediction distributions leads to suboptimal energy minimization, eroding prior classifier capabilities; 2) Non-target classifiers generate spurious low energy responses to current-stage data, violating distribution separation requirements.

To prevent the overall energy of certain classifiers from being lower than that of non-target, we design an energy

constraint mechanism that limits outputs with excessively low energy values. This ensures that classifiers from different stages maintain a reasonable distribution of energy values within the energy space when processing data from both target and non-target websites. Since the free energy is a negative value, we use its square as a loss function to penalize the low free energy values during output. The energy constraining loss function can be calculated by Eq. (8).

$$\mathcal{L}_F^t = \mathbb{E}_{x \sim P_D^t} \left(\mathbb{I}(F(x; f_{\theta}^t) < \zeta) \cdot F(x; f_{\theta}^t)^2 \right), \quad (8)$$

where ζ represents the threshold value for the triggering free energy constraining, and $\mathbb{I}$ denotes the discriminant function, the output of which is a boolean value. When $F(x; f_{\theta}^t)$ is less than the energy threshold ζ , the value is set to 1, thereby triggering the constraining to avoid over-sharp distributions. Conversely, if $F(x; f_{\theta}^t)$ is equal to or exceeds ζ , the value is set to 0, indicating that the constraining is not triggered.

Moreover, the training process of a neural network can be understood as optimizing the mutual information between data and model parameters. As free energy decreases, this mutual information increases, reflecting the model's improved capacity to capture data structure. However, classifiers trained at the current stage may still output high confidence scores for data from earlier stages, leading to false activations on out-of-distribution samples. This occurs because the current classifier θ_t may inadvertently encode feature patterns from historical data $\mathcal{D}_{t'}$, resulting in logit overlap, where predicted scores for target and non-target classes become indistinguishably close. As a consequence, the mutual information $I(\mathcal{D}_{t'}; \theta_t)$ becomes undesirably high.

To mitigate this, we aim to reduce the current classifier's ability to encode features from previous stages. Since the EIL-WF framework avoids retaining historical data, we cannot explicitly correct this behavior through data replay. Instead, we reduce mutual information indirectly by encouraging parameter distancing: the current classifier θ_t is encouraged to stay distant from historical classifiers $\{\theta_1, \dots, \theta_{t-1}\}$ in parameter space, thereby minimizing latent representational overlap. We implement this through the Parameter Distancing Mechanism, which seeks to reduce $I(\mathcal{D}_{t'}; \theta_t)$ for $t' \neq t$. The corresponding loss function term is:

$$\mathcal{L}_{\text{dis}}^{(t)} = \sum_{i=1}^{t-1} \|\theta_t - \theta_i\|_2. \quad (9)$$

This encourages the model to decouple from obsolete representations, reduces interference across stages, and promotes specialization on current tasks without relying on stored historical data.

The unified loss function combines these objectives:

$$\mathcal{L}_{\text{total}}^{(t)} = \mathcal{L}_{\text{CE}}^{(t)} + \lambda \mathcal{L}_{\text{energy}}^{(t)} + \gamma \mathcal{L}_{\text{dis}}^{(t)}, \quad (10)$$

where λ and γ balance the dual requirements of confident in-distribution classification and energy-space separation from non-target classifiers. When new stage data arrives, EIL-WF initializes a classifier and optimizes θ_t using $\mathcal{L}_{\text{total}}^{(t)}$, then freezes and aggregates θ_t with prior classifiers. This process is visualized in Algorithm 1 and the upper part of Figure 4.

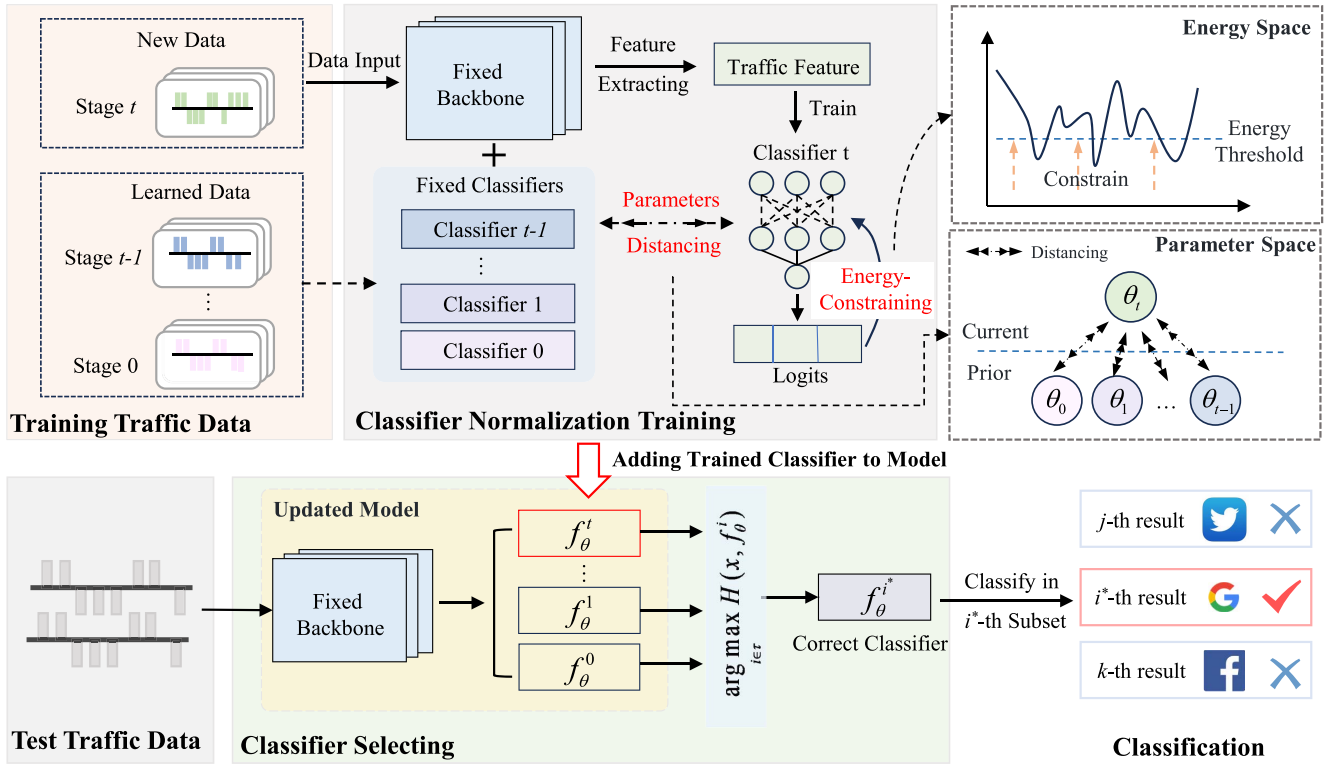


Fig. 4. Overall framework of the proposed EIL-WF. The upper part illustrates the process of classifier normalized training for the latest stage of the training phase, and the lower part depicts the classifier selecting process of multi-classifier aggregation during the testing phase. To aid comprehension, we provide schematic diagrams of two types of normalized training loss functions: energy constraining and parameter distancing.

C. Energy Alignment

Within the EIL-WF framework, classifiers operate independently, minimizing interference and isolating the model bias induced by imbalanced data. Furthermore, classifier normalization training constrains excessive energy outputs, ensuring minimal interference between model parameters. However, despite these advantages, EIL-WF encounters challenges in enhancing the performance of classifiers trained on low-sample data.

This limitation stems from practical constraints imposed by real-world network environments and security policies, which restrict the availability of low-sample data from certain websites. As a result, classifiers trained on limited or noisy data continue to exhibit suboptimal accuracy. This is particularly evident in the energy output space, where classifiers trained on low-sample data display significantly higher energy values compared to others. This discrepancy is illustrated in Figure 5(a), where the classifiers from stages 1 and 3, trained on low-sample data, exhibit notably higher energy outputs than those trained on Sufficient data. Consequently, classifiers trained on low-sample data may incorrectly classify test samples, as their disproportionately high energy values dominate the decision-making process despite the presence of more reliable classifiers.

To address this issue and ensure that classifiers trained on varying data qualities maintain comparable effectiveness during inference, we introduce the Energy Alignment (EA) strategy. EA is a lightweight and practical mechanism designed to harmonize the energy outputs of classifiers trained on

Algorithm 1 Classifier Normalization Training

Given components: Pre-trained backbone f_η , fixed stage classifier $\{f_\theta^i\}_{i=0}^{t-1}$, latest stage identification t , temperature T , learning rate ϵ ;
Input: Training data D_t ;
Output: The t -th classifier parameters θ_t ;

- 1: **for** x in D_t **do**
- 2: Initialize classifier f_θ^t for the stage t ;
- 3: Extract the traffic feature $f_\eta(x)$;
- 4: Generate the logits $f_\theta^t(f_\eta(x))$;
- 5: Calculate the cross-entropy loss L_{ce}^t by Eq. (6);
- 6: Calculate the free energy $F(x; f_\theta^t)$ by Eq. (3);
- 7: **if** $F(x; f_\theta^t) < \zeta$ **then**
- 8: Calculate $\mathcal{L}_F^t = F(x; f_\theta^t)^2$;
- 9: // Energy constraining loss
- 10: **end if**
- 11: Calculate $\mathcal{L}_\theta^t = \frac{1}{t} \sum_{i=0}^{t-1} \|\theta_t - \theta_i\|_2$;
- 12: // Parameters distancing loss
- 13: Calculate the total loss by $\mathcal{L}_{total}^t = \mathcal{L}_{ce}^t + \lambda \mathcal{L}_F^t + \gamma \mathcal{L}_\theta^t$;
- 14: Update θ_t by $\theta_t \leftarrow \theta_t - \epsilon \cdot \mathcal{L}_{total}^t$;
- 15: **end for**
- 16: **Return:** θ_t .

imbalanced datasets, especially those with severely limited training samples.

Rather than aligning the absolute energy values across classifiers, EA focuses on preserving a relative energy advantage

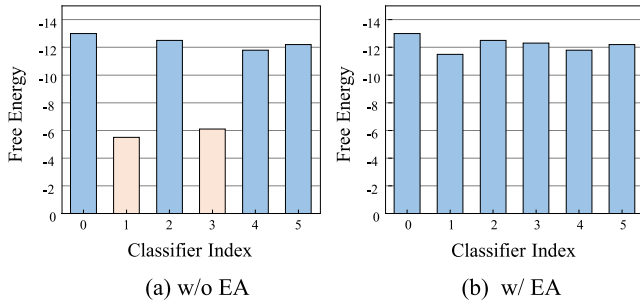


Fig. 5. Energy level of in-distribution data for stage classifiers with (b) and without (a) using EA where the 1st and 3rd classifiers are trained on low-sample data. The Y-axis represents the average energy output of the stage classifiers, and the X-axis represents the stage index of the classifiers.

for the target classifier. Specifically, the energy output of the classifier trained on low-sample data is adjusted by an additive scalar Δ , such that its total energy remains consistently lower than the energy output of non-target classifiers, with a safe margin defined by a tunable threshold ϵ . Formally, the condition is enforced as:

$$E_L + \Delta < E_{\text{non-target}} - \epsilon, \quad (11)$$

where E_L is the original energy output of the target classifier trained on low-sample data, $E_{\text{non-target}}$ denotes the energy values of all non-target classifiers for the same input, and ϵ is a small positive constant that prevents ambiguous overlaps due to numerical instability or noise.

This formulation ensures that, even in the presence of limited training data, the classifier corresponding to the target label remains energetically favorable compared to its non-target counterparts. As illustrated in Figure 5(b), the application of this adjustment results in energy outputs for low-sample classifiers that are not only more comparable to those from high-sample classifiers but also strategically prioritized during inference. The EA strategy, therefore, establishes a simple yet effective decision boundary in the energy space, mitigating the adverse impact of low data availability while enhancing prediction reliability.

To further enhance the performance of classifiers trained on low-sample data, we employ four data augmentation strategies: random insertion, random deletion, random splitting, and random merging. These techniques are designed to enrich the model's ability to capture the diverse features of website traffic. The random insertion and deletion strategies simulate fluctuations in packet size due to dynamic content changes [30], while the random merging and splitting strategies account for variations in burst size, resulting from differences in link bandwidths [13].

The EA strategy requires no modifications to the model's learning paradigm or the introduction of additional parameters. Instead, it applies a scalar adjustment during inference to improve the performance of classifiers trained on low-sample data. By aligning the energy outputs, we mitigate the negative effects of low-sample data and ensure that all classifiers, regardless of their training data scale, maintain similar energy levels, resulting in more accurate and robust predictions.

Algorithm 2 Classifier Selecting

Given components: The fixed backbone f_η , stage classifiers, $\{f_\theta^i\}_{i \in \tau}$, energy scalars $\Delta(i)$, $i \in \tau$;
Input: Test traffic sample x ;
Output: The classification result y^* ;

- 1: Extract traffic feature $f_\eta(x)$;
- 2: **for** $i = 1, \dots, t$ **do**
- 3: Generate the i -th logits $f_\theta^s(f_\eta(x))$;
- 4: Calculate the free energy $F(x; f_\theta^i)$;
- 5: Calculate the classification confidence $H(x, f_\theta^i)$
- 6: by Eq. (12);
- 7: **end for**
- 8: Selecting correct classifier identification i^* by Eq. (13);
- 9: Classify x in i^* -th subset by $y^* = \arg \max_{y \in Y^*} f_\theta^{i^*}(y | x)$;
- 10: **Return:** y^* .

D. Classifier Selecting

In previous studies [39], researchers show that data within the distribution generally have lower free energy compared to data outside the distribution, i.e., $F(x_i; f_\theta^i) > F(x_j; f_\theta^i), F(x_j; f_\theta^j) > F(x_i; f_\theta^j)$, where x_i represents in-distribution data of i -th stage, while j denotes out-of-distribution data of j -th stage.

During the training phase, we normalize the Helmholtz free energy outputs of each classifier, ensuring consistent comparability of the energy scale between classifiers at different stages. When training classifiers on low-sample datasets, we align their energy outputs with those trained on high-sample datasets by adding a shift scalar. These two strategies ensure that the energy outputs of classifiers at different stages remain consistent for in-distribution data, i.e., $F(x_i; f_\theta^i) \approx F(x_j; f_\theta^j), i \neq j$. Consequently, utilizing Helmholtz energy values during testing to select the most appropriate classifier for handling data represents a fair and reliable approach. Specifically, the confidence score is defined as the negative value of the Helmholtz free energy. For a given sample x , the classification confidence value of $H(x, f_\theta^i)$ can be defined as:

$$H(x, f_i) = -F(x; f_\theta^i) = T \log \sum_{y \in Y^i} e^{f_\theta^i(x, y)/T} - \Delta(i), \quad (12)$$

where $\Delta(i)$ represents the energy shift scalar of the i -th classifier output. If the classifier is trained on low-sample data, $\Delta(i)$ assumes a negative value; if it is trained on high-sample data, $\Delta(i) = 0$.

Then, the classifier with the lowest energy value is selected as the output result classifier (Eq. (13)).

$$i^* = \arg \max_{i \in \tau} \left(T \log \sum_{y \in Y^i} e^{f_\theta^i(x, y)/T} - \Delta(i) \right). \quad (13)$$

Finally, the output result of the selected classifier is taken as the traffic classification category by Eq. (5). The classifier selecting process is introduced in Algorithm 2 and presented in the lower part of Figure 4.

V. EXPERIMENTAL EVALUATION

In this section, we conduct a series of experiments to comprehensively evaluate the performance of the proposed EIL-WF method. We first outline the experimental setup, including dataset construction, evaluation metrics, model architecture, and training details in Section V-A.

We then assess the incremental learning capabilities of EIL-WF by adapting several state-of-the-art incremental learning algorithms to the website fingerprinting scenario and comparing their classification accuracy and forgetting rates with those of EIL-WF (Section V-B). Section V-C examines the adaptability of EIL-WF across different backbone networks and compares its consistency and robustness with other high-performing WF methods under multi-model structures. In Section V-D, we compare free energy based scoring and Softmax based scoring to evaluate their effectiveness in incremental identification tasks. We further examine real-world scenarios in Section V-E and Section V-F, focusing on performance under imbalanced dataset and concept drift dataset. Finally, Section V-G presents ablation studies to analyze the contribution of each loss function to overall incremental performance. In addition, the experimental environment of this work is based on a 4060 GPU with 12GB memory and PyTorch 2.1.1.

A. Experimental Setup

1) *Dataset*: To comprehensively verify the performance of EIL-WF, we conduct experiments on two classic public WF datasets, namely the DF95 dataset and the AWF100 dataset.

DF95: This dataset originates from [6], where Sirinam et al. collected 95 monitored websites using the TBB 7.X version. These websites are selected from the top 100 most popular sites listed by Alexa, with each website having 1000 traces. The sequence length of each sample is 5000.

AWF100: This dataset is sourced from [14], collected by Rimmer et al. using TBB V6.X. All websites are from the top of Alexa. In this paper, we use AWF100, which includes 100 monitored websites with non-intersecting URLs, each with 2500 traces, and each trace has a sequence length of 5000.

Drift dataset: To study the effect of distributional drift over time in the context of website fingerprinting, we introduce a series of temporally-shifted datasets derived from the original AWF100, which were also collected by Rimmer et al. These datasets, denoted as *AWF3d*, *AWF10d*, *AWF2W*, *AWF4W*, and *AWF6W*, correspond to traffic collected approximately 3 days, 10 days, 2 weeks, 4 weeks, and 6 weeks after the original AWF100 collection period, respectively. Each class contains 100 samples, and each sample has a trace length of 5000.

Incremental data setting: In the initial phase of the incremental learning model, the number of categories comprises approximately half of the total categories in the dataset, with ten new website categories added in each subsequent incremental stage. In the DF95 dataset (comprising 95 categories), the initial phase includes 45 categories; in the AWF100 dataset (comprising 100 categories), the initial phase includes 50 website categories. The dataset is divided into training, validation, and test sets in an 8:1:1 ratio. For example, for each website in

the DF95 dataset, which contains 1,000 samples, 800 samples are allocated for training, while both the validation and test sets comprise 100 samples per website.

2) *Evaluation Metrics*: Considering that the primary objective of website fingerprinting is to accurately identify websites and avoid mis-classification, average accuracy (AAC) is designated as a crucial evaluation metric. The purpose of incremental learning is to reduce the forgetting rate of old categories while continuously incorporating new categories. Final forgetting rate (*FF*) serves as a measure to assess the incremental capability of EIL-WF, which quantifies the degree of performance degradation of old tasks or categories after the introduction of new tasks or categories. Given that the classification accuracy within the classifier relies on the performance of the selected backbone network, the classification accuracy of EIL-WF is principally influenced by the ability to classify the samples accurately within the correct classifier. We employ average recall (*AR*) to demonstrate the capability of EIL-WF in selecting the appropriate classifier for test data. *AAC*, *AR*, and *FF* can be computed as follows:

$$\begin{aligned} AAC &= \frac{\sum_{i=1}^t N_i \cdot ACC_i}{N_{total}} \\ AR &= \frac{\sum_{i=1}^t N_i \cdot Recall_i}{N_{total}} \\ FF &= AAC_{UB}^{old} - AAC_{new}^{old}, \end{aligned} \quad (14)$$

where ACC_i and $Recall_i$ denote the accuracy and recall rate of each classifier for in-distribution data, N_i represents the sample size of the i -th classifier, and N_{total} indicates the total sample size of all current websites. AAC_{UB}^{old} is the upper bound average accuracy of the old data (the accuracy of the model trained with the complete old data), while AAC_{new}^{old} reflects the average accuracy of the old data following the model update.

3) *Parameter Setting*: The temperature T is set to 0.4, and the regularization coefficients for the energy constraining loss $\lambda = 0.25$ and the parameter distancing loss $\gamma = 0.08$ are determined. EIL-WF uses *DFnet* as the default backbone network structure for benchmark testing and maintains consistency with its network hyperparameters.

B. Comparison With CIL Methods in WF

In this part, we compare the incremental learning capacity of EIL-WF with existing CIL methods in the WF scenario.

To enhance clarity, we adapt and apply classic CIL frameworks *iCarl* [15], *Co2L* [33], *L2P* [35], *EASE* [16], *APER* [38] and *ESN* [34] to WF and conduct a comparison. *iCarl* and *Co2L* represent incremental learning methods that require the retention of 50 samples per historical category for model training. *L2P*, *EASE*, *APER* and *ESN* are incremental learning methods that do not require caching old data.

iCarl [15]: learns new classes incrementally by jointly learning representations and classifiers while maintaining a small memory buffer.

Co2L [33]: employs contrastive learning and self-supervised distillation to improve representation robustness and transferability.

TABLE I

COMPARATIVE ANALYSIS OF AVERAGE ACCURACY ($\uparrow$) AND FINAL FORGETTING ($\downarrow$) BETWEEN EIL-WF AND EXISTING INCREMENTAL LEARNING METHODS ON DF95 AND AWF100 DATASETS. BUFFER SIZE REPRESENTS THE NUMBER OF SAMPLES CACHED FOR EACH OLD WEBSITE, t DENOTES THE LATEST STAGE NUMBER. ALL NUMBERS ARE %, AND WE DO NOT SHOW STANDARD DEVIATIONS LESS THAN 0.5%

Method	Buffer Size	Dataset Stage	AWF100					DF95				
			t=1	t=2	t=3	t=4	t=5	t=1	t=2	t=3	t=4	t=5
UB	—	AAC	99.5	99.5	99.4	99.4	99.3	98.6	98.4	98.3	98.2	98.1
iCarl[15]		AAC	97.5	91.3 $\pm$ 0.5	88.2	82.2	76.4 $\pm$ 2.4	94.5	87.5 $\pm$ 0.6	80.2	75.2 $\pm$ 1.6	70.4 $\pm$ 2.2
		FF	1.2	7.1	9.6	14.9	21.7 $\pm$ 1.7	3.6	9.1 $\pm$ 0.5	16.2	21.8 $\pm$ 1.2	26.5 $\pm$ 1.4
Co2L[33]	50/class	AAC	97.8	94.3	91.4	87.2	84.3	96.2	92.4 $\pm$ 0.5	88.3	81.4	75.2
		FF	0.8	3.3	6.8	11.4	13.1	1.6	5.5 $\pm$ 0.7	8.7	16.3	22.4
L2P[35]		AAC	98.5	93.1	90.2	83.2 $\pm$ 1.6	79.5 $\pm$ 3.2	96.3	91.2	86.2	78.4 $\pm$ 0.9	73.5 $\pm$ 0.7
		FF	0.3	5.4	7.3	14.8 $\pm$ 1.4	18.1 $\pm$ 2.6	1.4	5.3	8.2	15.8 $\pm$ 0.5	23.6 $\pm$ 0.5
EASE[16]		AAC	98.3	94.1	90.2	88.5	85.4 $\pm$ 0.6	96.4	94.1	90.3	88.2 $\pm$ 0.6	85.6 $\pm$ 0.7
		FF	1.4	3.6	8.8	9.7	10.8	1.6	3.4	6.5	9.3	11.4
ARER[38]	0	AAC	97.5	93.3	93.2	87.4	83.5 $\pm$ 0.6	95.5	91.3	86.6	84.3 $\pm$ 1.5	82.4 $\pm$ 0.8
		FF	1.1	5.7	6.1	10.2	12.4	2.1	6.9	9.8	12.1 $\pm$ 1.1	14.2
ESN[34]		AAC	99.1	99.0	94.7	92.2 $\pm$ 0.6	90.4	98.4	96.5	93.3	91.2	88.4
		FF	0.2	0.2	4.1	5.3	7.1	0.1	1.5	4.1	5.8	8.4
EWF(ours)		AAC	99.3	98.8	97.7	97.2	96.6	98.3	97.1	96.9	96.3	95.3
		FF	0.1	0.3	1.1	1.3	1.7	0.2	0.5	0.5	1.3	2.1

L2P [35]: L2Puses prompt-based tuning of pre-trained models, storing prompts as reusable memory for future tasks.

EASE [16]: EASE builds a scalable subspace via lightweight adapters on frozen pre-trained models, leveraging intra-class similarity in a collaborative space to synthesize classifiers for old classes without accessing past data.

APER [38]: APER balances generalization and adaptability by efficiently fine-tuning pre-trained models and combining their features with adapter outputs.

ESN [34]: ESN expands its classifier set dynamically and aligns their outputs via energy-based self-normalization and ensemble inference.

The comparison results are presented in Table I. To facilitate a more effective comparison, this study introduces the concept of an upper bound (UB) at each stage, representing the accuracy of training the model with the complete dataset at each stage. The results demonstrate that EIL-WF, which does not require caching historical data on the same traffic feature extraction network, achieves superior performance in most cases.

Over the AWF100 dataset, the final *AAC* of EIL-WF reaches 96.6%, which is only 2.7% lower than the UB. Compared to the best method among those relying on data replay, the *AAC* improves by 19.8%, and EIL-WF outperforms Co2L in *AAC* at all stages. In comparison to methods that do not require caching, the final *AAC* of EIL-WF is 6.2% higher than the best method, ESN. Although the performance remains similar in the initial incremental stage and is slightly lower than ESN at the second stage, as the number of stages increases, the classification interference between stages in ESN intensifies, leading to a rapid decline in its performance. After the number of stages t exceeds 2, EIL-WF consistently achieves the best results.

On the DF95 dataset, the final *AAC* of EIL-WF reaches 95.3%, which is 2.8% lower than the UB and 20.1% higher than the best method among those relying on data replay in incremental learning approaches. EIL-WF also outperforms Co2L in *AAC* at all stages. Compared to methods that do not require caching, the final *AAC* of EIL-WF is 6.9% higher than the best method, ESN. Although it is slightly lower than ESN at stage $t = 1$, EIL-WF consistently performs better than ESN in subsequent incremental stages, and its advantage becomes more pronounced as the number of stages increases. Due to the isolation between classifiers at each stage in EIL-WF, the interference between new and old data remains minimal, resulting in a lower forgetting rate for old data. Compared to other methods, EIL-WF achieves the best or similar performance. On the AWF100 dataset, the final forgetting rate of EIL-WF is 1.7%, which is 20% lower than iCarl, 11.4% lower than Co2L, 16.4% lower than L2P, and 5.4% lower than ESN. On the DF95 dataset, the final forgetting rate of EIL-WF is 2.1%, which is 5.7% lower than the best method, ESN, and 20.4% lower than the average of the four compared methods. This indicates that the method proposed in this paper provides continuous learning capabilities for WF models, meeting the needs of long-term dynamic updates to the monitored set.

C. Comparison With Existing WF Methods Under CIL Setting

The EIL-WF is a general incremental learning framework designed for website fingerprinting, aiming to equip the majority of WF models with the capability for continuous learning. To assess the effectiveness of EIL-WF in enhancing incremental learning performance for website

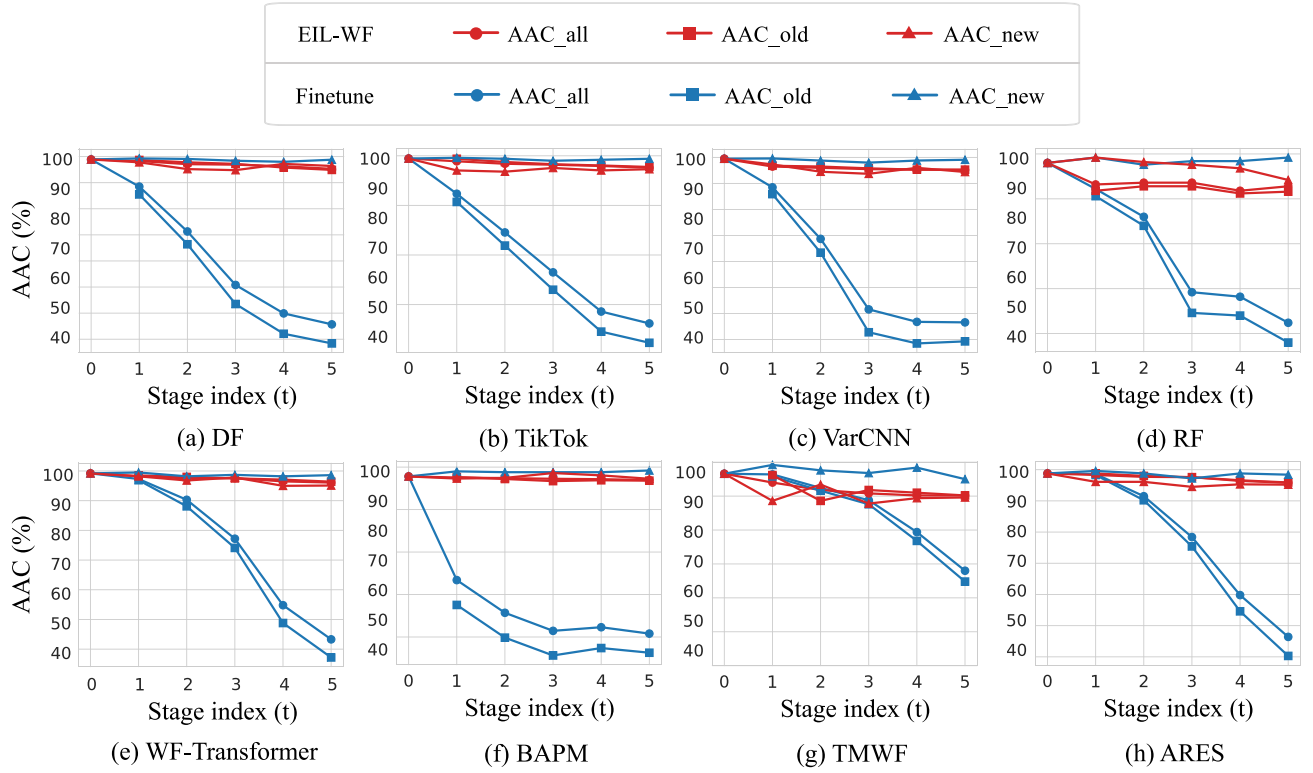


Fig. 6. Comparison of classification accuracy across different WF models using EIL-WF and finetune strategies over five incremental learning stages. The metrics include average accuracy over all classes (AAC_{all}), average accuracy over old classes learned in previous stages (AAC_{old}), and average accuracy over newly introduced classes in the current stage (AAC_{new}).

fingerprinting tasks, we design a controlled comparative experiment. Specifically, eight representative WF models are selected as fixed backbone architectures: DF [6], TikTok [7], VarCNN [8], RF [9], WF-Transformer [10], BAPM [28], TMWF [11], and ARES [12]. Among these, BAPM, TMWF and ARES are multi-tab WF methods. For consistency, we configure all models with a single tab during experimentation. We establish a baseline method for comparison: a standard fine-tuning-based incremental learning strategy, in which the backbone network remains frozen during training, and a dynamically decaying learning rate is applied to alleviate catastrophic forgetting. Furthermore, to highlight the efficiency of EIL-WF, we also compare its computational and time costs with those incurred by a full retraining strategy, where the model is retrained from scratch on the entire accumulated dataset at each incremental stage.

To evaluate the incremental learning performance of each method, we adopt three metrics: average accuracy over all classes (AAC_{all}), average accuracy over old classes (AAC_{old}), and average accuracy over new classes (AAC_{new}). These metrics are computed for each approach across five incremental stages, from $t = 1$ to $t = 5$. All experiments are conducted on the DF95 dataset.

The experimental results, illustrated in Figure 6, provide strong empirical evidence for the effectiveness of EIL-WF across diverse WF backbone models. The framework consistently achieves high classification accuracy across all

evaluation metrics, demonstrating both robustness and adaptability in incremental learning scenarios. After completing five incremental learning stages, EIL-WF maintains high overall accuracy across all tested architectures: DF, TikTok, VarCNN, RF, WF-Transformer, BAPM, TMWF, and ARES were 95.3%, 95.2%, 95.2%, 96.4%, 96.1%, 93.9%, 95.1%, and 95.8%, respectively. This consistency underscores the framework's ability to mitigate the negative effects of distributional shifts and class expansion.

In stark contrast, traditional fine-tuning approaches show a marked decline in overall performance, particularly in AAC_{all} , as the learning progresses through incremental stages. At stage five, DF, TikTok, VarCNN, RF, WF-Transformer, BAPM, TMWF, and ARES dropped to 35.7%, 32.4%, 36.6%, 81.2%, 43.3%, 21.6%, 84%, and 46.4%, respectively—highlighting their vulnerability to catastrophic forgetting. This sharp performance drop is primarily due to the erosion of learned representations for earlier tasks, which are not adequately preserved during subsequent updates. Notably, while fine-tuned models may demonstrate reasonable performance on newly introduced classes (reflected in initially elevated AAC_{new} scores), they fail to retain recognition capabilities for previously learned classes, as evidenced by rapidly declining AAC_{old} scores. In contrast, EIL-WF successfully balances plasticity and stability, allowing models to acquire new knowledge while maintaining competence on earlier tasks. This capability is particularly critical for real-world deployment of WF systems, where continuous updates

TABLE II
TIME OVERHEAD OF WF MODELS FOR MODEL UPDATING WITH AND WITHOUT EIL-WF FRAMEWORK, MEASURED IN SECONDS(S)

Method	Computing time	
	w/o EIL-WF	w/ EIL-WF
DF [6]	1.1×10^3	5.2
TikTok [7]	1.3×10^3	5.4
VarCNN [8]	5.5×10^3	34.5
RF [9]	1.8×10^3	9.5
WF-Transformer [10]	1.2×10^5	4.1×10^3
BAPM [28]	7.2×10^2	3.1
TMWF [11]	1.5×10^3	7.6
ARES [12]	2.3×10^3	10.2

are necessary without compromising historical classification integrity.

In the incremental learning phase, EIL-WF utilizes only the data from the current stage to train the classifier, thereby facilitating rapid model updates. This approach proves particularly advantageous for WF scenarios characterized by limited storage and computational resources. Table II presents a comparison of computational time costs between traditional methods and EIL-WF during model updates. The results demonstrate that the training time of EIL-WF significantly reduces in comparison to traditional methods, with a decrease of $2 \sim 3$ orders of magnitude. This advantage arises from the necessity of traditional methods to employ the entire dataset for training the model from scratch, which results in gradually increasing time costs. In contrast, EIL-WF maintains stability and reduces training time consumption throughout the process by utilizing only the current data for training.

Furthermore, EIL-WF training eliminates the need for caching historical data, significantly reducing the storage space required for model training. As the number of incremental stages increases, traditional methods demand the continuous retention of an expanding volume of original website category data, leading to a substantial rise in storage costs. For instance, when starting with 45 website categories and adding 10 new categories at each incremental stage, EIL-WF retains only the data relevant to the current classifier's training, which can be deleted after training. This process ensures that, at any given stage, the model stores data for up to 10 website categories, thereby maintaining low overall storage consumption. In contrast, traditional methods must store all data, resulting in escalating storage costs over time. The efficiency of EIL-WF in utilizing computational and storage resources provides it with a distinct advantage in long-term tasks.

D. Comparison With Softmax-Based Classifier Selecting

This section evaluates the effectiveness of the proposed free energy-based scoring mechanism as a confidence measure for classifier selection during inference. As the number of classifiers increases across incremental stages, this mechanism helps route each test sample to the most suitable classifier, thereby maximizing classification accuracy. It plays a critical role in the EIL-WF framework by preventing catastrophic forgetting and supporting continual learning. We compare

this approach with the conventional Softmax-based confidence measure, which serves as the baseline. To visualize how well each method performs in classifier selection, we conduct experiments on the DF95 dataset and test at every incremental stage from $t = 1$ to $t = 5$. Figure 7 presents confusion matrices that illustrate classifier assignment performance. Diagonal elements represent correctly routed samples, while off-diagonal elements indicate misrouted samples, revealing inter-classifier interference.

The results clearly demonstrate that the free energy-based scoring strategy significantly outperforms the conventional Softmax-based approach in selecting the correct classifier during incremental inference. Under the free energy strategy, the confusion matrices exhibit strong diagonal dominance, indicating accurate sample-to-classifier routing and minimal interference between classifiers. In contrast, the Softmax method leads to increased off-diagonal errors and substantial cross-classifier confusion, which intensifies with each incremental stage. At stage $t = 5$, the Softmax-based approach shows a notable decline in recall, with average per-classifier recall dropping to 47.9%, 86.0%, 85.1%, 92.3%, 92.5%, and 88.1%, respectively.

By contrast, the free energy-based method consistently maintains high recall, successfully directing the majority of test samples to their appropriate classifiers. At the same stage ($t = 5$), the average recall across classifiers remains robust at 99.0%, 97.7%, 97.4%, 96.8%, and 96.1%. These findings confirm that the proposed energy-based scoring mechanism not only improves routing accuracy and overall classification performance but also effectively suppresses classifier interference in dynamic, multi-stage incremental learning scenarios.

E. Performance on Imbalanced Dataset

In this part, we evaluate the performance of EIL-WF on imbalanced datasets. Since existing WF methods are not designed to address class imbalance issues and few-shot methods typically exhibit high spatial complexity, we employ the original DF model as the baseline for comparative analysis with the EIL-WF method on imbalanced datasets. Specifically, EIL-WF is categorized into two settings: one with energy alignment (EA) and the other without, to illustrate the applicability of EIL-WF in addressing imbalanced datasets and to assess the effect of energy alignment. We randomly select 20 categories from the DF95 and AWF100 datasets as low-sample website categories, which contain fewer samples $N \sim (2, 20)$, to simulate dataset imbalance. We use L -ACC to represent the average accuracy of low-sample websites during testing.

The experimental results, as shown in Figure 8, demonstrate the comparative outcomes under varying degrees of data imbalance. It is observed that EIL-WF with energy alignment outperforms the baseline method when the number of samples for poorly performing websites is fewer than 20. In the case of $N = 2$, the initial method's recognition AAC for these websites is 0, indicating that it is challenging to learn about these websites due to model training bias. However, EIL-WF (with EA) achieves a recognition AAC of 78.2% on the

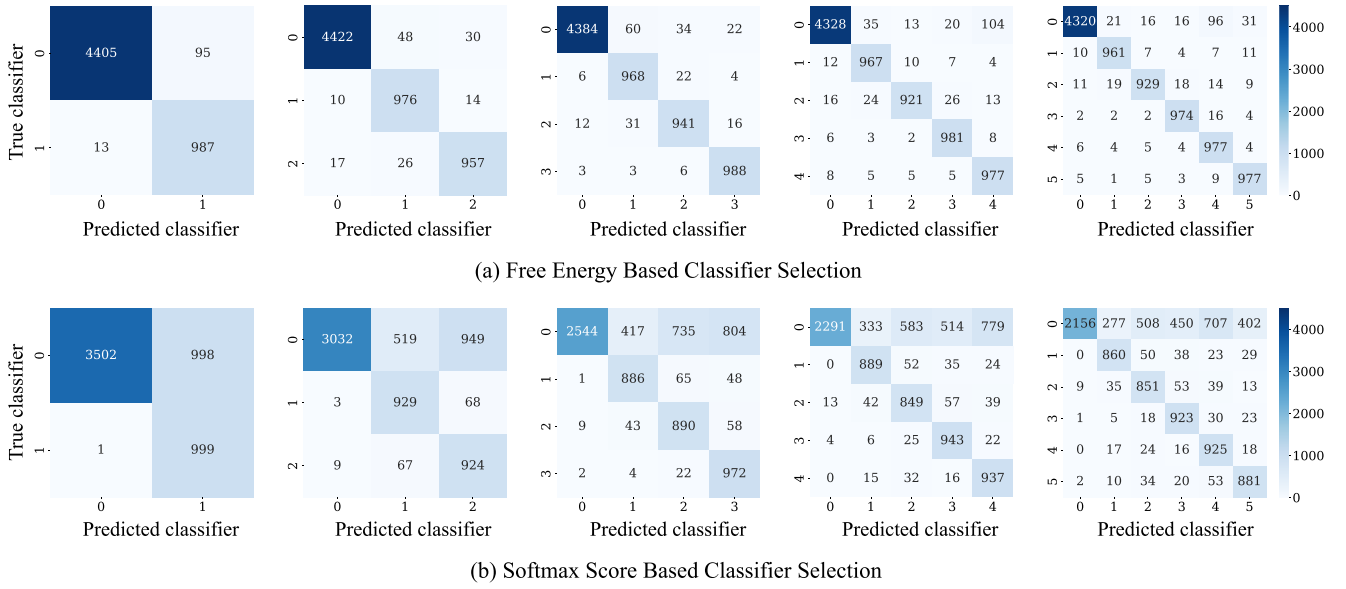


Fig. 7. Confusion matrices comparing classifier selection performance using free energy-based versus Softmax-based scoring mechanisms across incremental stages ($t = 1$ to $t = 5$) on the DF95 dataset.

DF95 dataset and over 70% on the AWF100 dataset for these websites. Notably, EIL-WF without EA also recognizes these poorly performing websites to some extent.

When the number of low-sample website examples $N = 20$ increases, the model bias issue of traditional methods is somewhat mitigated, with recognition AAC reaching approximately 71.7% on the DF95 dataset and 77.5% on the AWF100 dataset for these websites, which remains below the ideal level. In contrast, EIL-WF (with EA) significantly improves recognition performance, with AAC reaching 92.4% on the DF95 dataset and 92.1% on the AWF100 dataset, demonstrating its strong ability to recognize websites with low-sample samples in the presence of data imbalance.

To improve the principled and practical determination of the energy scalar Δ in automated or large-scale deployments, we conducted an in-depth analysis of its calculation. The primary objective is to ensure that the majority of traffic from low-sample websites can be correctly classified. As discussed in Section IV-C, this requires that during inference, the energy output of the target classifier remains consistently lower than that of the non-target classifiers, with a sufficient margin defined by the threshold ϵ . The choice of ϵ therefore directly determines the extent to which the target classifier's energy must be reduced. Specifically, we compute the average energy outputs for low-sample website data across both the target and non-target classifiers, introduce an additive scalar to align these outputs, and then determine an appropriate ϵ to maintain the target classifier's relative advantage.

To systematically analyze the impact of different ϵ values on classification performance, we explored a range of ϵ values within $[-2, 0]$ and observed that when ϵ lies within the interval $[-0.8, -0.4]$, the overall classification accuracy remains relatively stable across various sample size settings. As the absolute value of ϵ increases (i.e., becomes more negative), the classification accuracy for low-sample websites steadily improves. However, if the energy adjustment is too

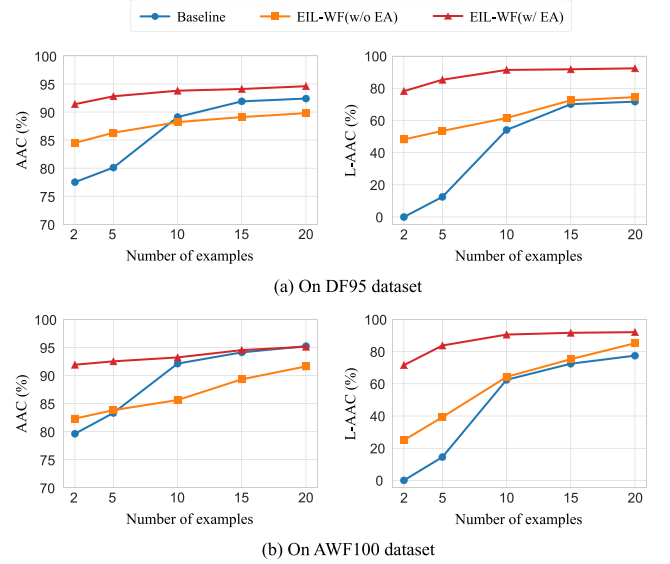


Fig. 8. Average accuracy performance on imbalanced data, with the X-axis representing the sample numbers of 20 low-sample data site categories. (a) shows the results on the DF95 dataset, and (b) shows the results on the AWF100 dataset.

aggressive, it introduces excessive interference to other classifiers, resulting in a noticeable decline in overall system accuracy. To adopt a conservative approach and minimize potential negative impact on other stages, we ultimately set $\epsilon = -0.5$ as the default configuration for this experiment.

F. Performance on Concept Drifted Dataset

This section evaluates the performance of EIL-WF in handling concept drift in dynamic environments. Concept drift refers to changes in data distributions over time that lead to model performance degradation. Traditional models trained on static datasets often fail to adapt to such changes. Common

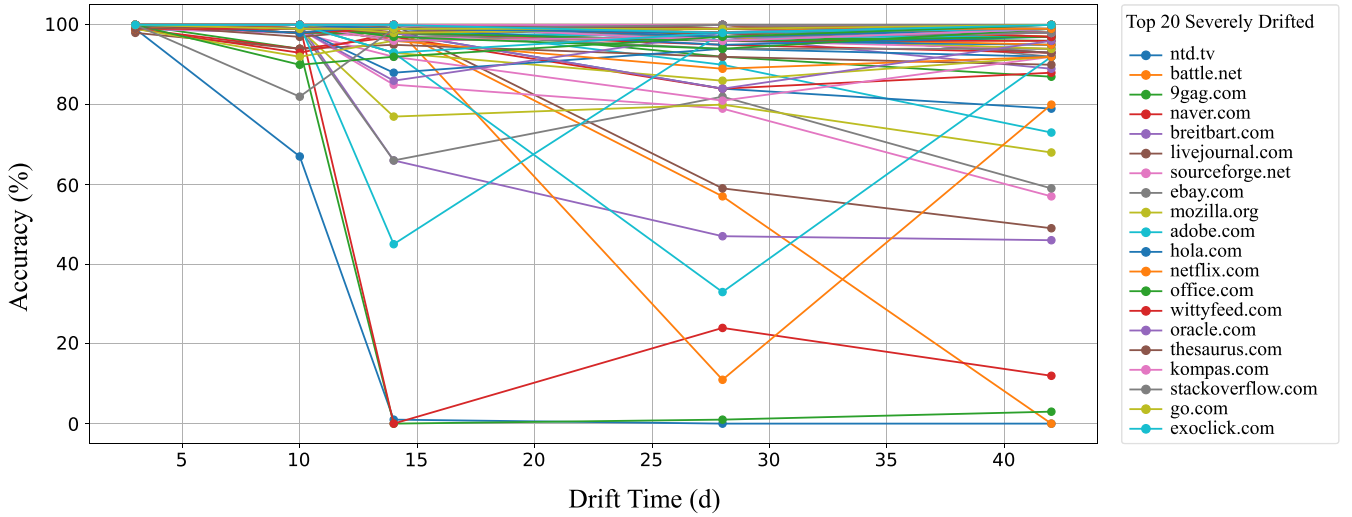


Fig. 9. Impact of concept drift on WF classification accuracy over time. The model is initially trained on the AWF100 dataset and evaluated on multiple temporally drifted datasets (AWF3d, AWF10d, AWF2w, AWF4w, and AWF6w).

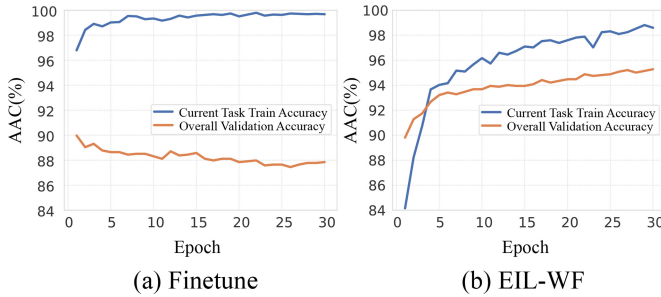


Fig. 10. Comparison of classification accuracy between EIL-WF and fine-tuning methods on the top 15 severely drifted website categories, showing both current task accuracy for updated categories and overall accuracy.

strategies for addressing this issue rely on full data recollection and model retraining at regular intervals. These approaches are rarely feasible in real-world deployment due to high demands on storage, communication, and computation.

To evaluate the impact of concept drift on model performance, we train an initial model based on DFnet using the AWF100 dataset, and test it on several drifted versions of the dataset: AWF3d, AWF10d, AWF2w, AWF4w, and AWF6w. As shown in Figure 9, the model's classification accuracy is significantly affected by concept drift, dropping from an initial 99.1% to 88.7% after six weeks. However, it is worth noting that this performance degradation does not occur uniformly across all classes. On the AWF6w dataset, only 15 website classes exhibit classification accuracy below 90%, while the majority of classes maintain relatively stable performance. This suggests that in practice, concept drift tends to be non homogeneous and affects only a subset of high dynamic websites within a given time window.

A more practical and cost effective solution involves periodically sampling small data subsets and selectively updating classifiers for high drift categories. This approach preserves system adaptability without requiring full model retraining. EIL-WF performs well under this strategy. Its modular multi

classifier architecture allows each classifier to operate independently. Updating one does not interfere with the others, which enables precise incremental learning. In contrast, fine tuning improves recognition for selected categories but often causes forgetting of stable ones. This problem is especially severe under class imbalance, where accuracy for untrained classes may degrade significantly and compromise system robustness.

We trained the DFnet model on AWF100 and then selected the 15 most high drift categories from AWF6w. Each website provided 80 training samples and 10 validation samples. Final testing was conducted on the complete AWF6w dataset. In the fine tuning experiments, we froze convolutional layers and lowered the learning rate to limit gradient changes. In EIL-WF, the selected categories were assigned to a separate classifier that was trained independently while the remaining classifiers remained unchanged.

Figure 10 presents the final results. Fine tuning raised the AAC of selected categories to 99.1% but caused a drop in overall performance. After 30 epochs of training, overall AAC decreased to 87.8%. EIL-WF, on the other hand, improved AAC for the 15 selected categories to 98.2% while maintaining stable performance for the rest. Overall AAC reached 95.2%. These results show that EIL-WF not only improves recognition for high drift categories but also avoids disrupting stable ones.

G. Impact of Loss Function Configurations

To analyze the specific impact of the two loss functions in classifier normalization training on the performance of the enhanced learning method, we conduct ablation experiments. These experiments evaluate the effect of each loss function individually or in combination on the final performance of the model.

As shown in Table III, when the model is trained solely with cross-entropy loss, the overall performance remains suboptimal, with AAC rates of 88.1% and 89.2% on the DF95 and AWF100 datasets, respectively. This is accompanied by elevated forgetting rates, reaching 8.9% and 9.1%. The error in

TABLE III
IMPACT OF EACH LOSS FUNCTION ON AAC AND AR,
ALL NUMBERS ARE %

Loss function	DF95		AWF100	
	AAC	AR	AAC	AR
w/o $\mathcal{L}_\theta$	93.3	94.3	94.2	94.8
w/o $\mathcal{L}_{ec}$	91.6	92.2	92.1	92.6
w/o $\mathcal{L}_\theta, \mathcal{L}_{ec}$	88.1	89.1	89.2	89.7
None	95.3	96.9	96.6	97.1

selecting data for classifiers is also considerable, with average recall rates of only 89.1% and 89.7% for each classifier. This indicates that in the absence of additional loss functions, as the number of learning stages increases, the model's stages become more susceptible to interference during classification, leading to a more pronounced catastrophic forgetting problem. In other words, the acquisition of new knowledge significantly undermines the retention of prior knowledge.

Further analysis reveals that the introduction of the energy constraining loss effectively alleviates this issue. Without this loss function, the average recall rate for each classifier on the two datasets is 92.2% and 92.6%, respectively, indicating a significant decline compared to EIL-WF. This results in a reduction in *AAC* rates by 3.7% and 4.5%, respectively. This demonstrates the important role of the energy constraining loss in reducing interference between classifiers at different stages and maintaining long-term memory.

Additionally, the results show that the parameter distance loss contributes to improving the model's incremental learning ability. By encouraging the model parameters to maintain a distance from the parameters of specific classifiers, it reduces the incorrect responses of non-target classifiers to new data. The model without this loss has final *AAC* rates of 93.3% and 94.2%, which are 2.3% and 2.4% lower than those obtained with the complete loss setting, indicating its positive role in optimizing model generalization and preventing overfitting.

H. Discussion

This study proposes an incremental learning approach for website fingerprinting attacks based on a multi-classifier architecture. The method enables traditional WF attack models to learn continuously, overcoming their reliance on the assumption of static website categories. It provides structural support for adapting WF models to dynamic network environments and shows strong scalability and flexibility, particularly when facing frequent category changes or rapid content updates.

Within the incremental learning framework, we introduce mechanisms such as classifier-normalized training to align the confidence output distributions of newly added classifiers across stages. This mitigates unfair inference caused by distribution shifts between stages and reduces bias when aggregating outputs from multiple classifiers. The design improves overall model stability and generalization. However, although each newly added classifier is designed to be lightweight and accounts for approximately 1.6% of the parameters of the backbone network, the model still faces scalability challenges in practical deployment. As incremental

stages accumulate, the number of classifiers may grow to dozens or even hundreds, resulting in a significant increase in the total number of parameters and placing a considerable burden on resource-constrained edge devices. To address this, future work may explore model compression strategies such as model distillation, which merges stage-wise classifiers into a unified model, or structured pruning to eliminate redundant parameters. These techniques can help control model size without compromising performance, enhancing deployability.

We also introduce an energy alignment (EA) strategy to calibrate output energies across classifiers trained with imbalanced data. This effectively reduces performance discrepancies caused by varying training sample sizes across stages. Experimental results confirm that EA significantly improves performance in few-shot stages and offers a clear and efficient solution to data imbalance in incremental WF learning. However, the current EA approach focuses on inter-stage imbalance and does not fully account for intra-stage distribution disparities. In reality, variations in sample quantity within a single stage may also introduce training bias and affect fine-grained classification. Future research could incorporate sample-level energy scaling or integrate established techniques such as focal loss or class reweighting to better address intra-stage imbalance.

From an architectural perspective, the proposed multi-classifier design supports localized updates, allowing targeted optimization for specific domains without affecting other components. This capability offers a practical solution for addressing concept drift in WF attacks. When website structures evolve over time, lightweight classifiers can be trained and deployed for affected sites, preserving adaptability and monitoring accuracy. Compared to full retraining, localized updates require fewer resources and offer faster responsiveness. A promising future direction is the development of feature drift detection mechanisms. For instance, designing lightweight drift-aware models that detect subtle structural or behavioral changes could guide classifier updates and support more precise and dynamic incremental learning.

VI. CONCLUSION

This paper presents an incremental learning approach named EIL-WF, enabling website fingerprinting models to evolve continuously as new websites emerge. By training lightweight classifiers to learn new knowledge and incorporating classifier-normalized training with a free energy-based selection mechanism, the framework effectively integrates multi-stage classifiers while preventing forgetting of previously learned information. In addition, parameter-isolated training alleviates bias caused by data imbalance and supports localized adaptation under concept drift. Experimental results demonstrate that EIL-WF achieves strong accuracy, adaptability, and efficiency in dynamic environments, making it well-suited for deployment in real-world continual learning scenarios.

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